

Milestone Optimization for Brand Campaigns

Data Analyst Take-Home Assignment

A data-driven approach to recommending milestone ladders using historical campaign performance.

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Contents

1	Methodology	2
1.1	Definition of Optimized	2
1.2	Recommendation Approach	2
1.3	Payout Allocation	2
1.4	Handling New Campaigns	2
1.5	Assumptions and Limitations	3
2	Stakeholder Summary	3
3	Results	4
3.1	Example Recommendation	4
3.2	Backtesting	4

1 Methodology

1.1 Definition of Optimized

For this project, an *optimized* milestone ladder is one that balances creator motivation, fairness, and budget control. The goal is to recommend milestone targets that are realistic for similar campaigns while ensuring the total payout never exceeds the allocated campaign budget. Rather than maximizing a single metric, the approach aims to create milestones that are achievable, progressive, and consistent across campaigns.

1.2 Recommendation Approach

The recommendation process begins by combining campaign, creator, post, and historical milestone datasets into a single analytical dataset. Historical campaigns are grouped by **campaign category** and **creator tier**, as these are the strongest indicators of expected campaign performance.

For each group, the distribution of historical final view counts is analysed. Milestone thresholds are then placed at the 20th, 40th, 60th, 80th, and 90th percentiles of the distribution. Using percentiles creates milestones that increase gradually while remaining realistic for campaigns with similar characteristics, with the 90th percentile serving as a practical upper bound to prevent setting unreachable targets.

1.3 Payout Allocation

After the milestone thresholds are generated, the campaign budget is distributed across the five milestones using fixed percentages.

Milestone	Budget Allocation
1	10%
2	15%
3	20%
4	25%
5	30%

Table 1: Budget allocation across milestones

This structure rewards creators for early progress while providing larger incentives for stronger performance, without exceeding the campaign budget.

1.4 Handling New Campaigns

Some campaigns may not closely match historical data. In these situations, the recommendation follows a fallback strategy.

- Use campaigns with the same category and creator tier whenever available.
- If insufficient data exists, use the average for that campaign category.
- If category-level information is also unavailable, use the overall historical average.

This ensures that every campaign receives a recommendation, even when historical examples are limited.

1.5 Assumptions and Limitations

The approach assumes that historical campaign performance is a reasonable indicator of future performance and that campaign category and creator tier are meaningful predictors of expected views.

The model does not account for factors such as content quality, seasonal trends, audience engagement, or platform algorithm changes. Recommendations for campaigns that differ significantly from historical patterns may therefore be less accurate. Additional data, such as engagement metrics or audience demographics, would help improve future recommendations.

2 Stakeholder Summary

Why was this built?

Planning milestone ladders manually can lead to inconsistent targets and requires significant effort. This solution recommends milestone targets automatically using historical campaign data, making the process faster, more consistent, and easier to justify.

How does it work?

When a new campaign is created, the system looks at previous campaigns with a similar category and creator tier. It estimates a realistic range of expected views and recommends five milestone targets. The campaign budget is then distributed across these milestones while ensuring that the total payout remains within budget.

Why is this better?

This approach replaces guesswork with historical evidence. It provides:

- Consistent milestone recommendations.
- Fair targets for different creator tiers.
- Budget-compliant payout structures.
- Less manual effort during campaign planning.

Because every recommendation follows the same process, decisions are easier to explain and remain consistent across future campaigns.

When should it be used with caution?

The recommendations are based on historical campaign performance. If a campaign is very different from previous campaigns or is affected by factors such as viral trends or platform algorithm changes, the recommended milestones may require manual review.

The system is designed to support decision-making rather than replace it. As more campaign data becomes available, the recommendations can be updated and become increasingly accurate.

3 Results

The proposed methodology was evaluated by generating milestone recommendations for a sample campaign and comparing the recommendations with historical milestone configurations.

3.1 Example Recommendation

Figure 1 shows the milestone ladder generated for a Gaming campaign targeting Mid-tier creators with a campaign budget of 400,000. The recommended thresholds increase progressively while distributing the budget across five milestones. The total payout remains equal to the allocated campaign budget.

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===== Test a New Campaign =====
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Available Categories:

1. Beauty
2. Fashion
3. Fitness
4. Food
5. Gaming
6. Tech

Available Creator Tiers:

1. Macro
2. Micro
3. Mid
4. Nano

	Milestone	View Threshold	Payout
0	Milestone 1	6,662	40,000
1	Milestone 2	15,704	60,000
2	Milestone 3	24,552	80,000
3	Milestone 4	56,192	100,000
4	Milestone 5	238,904	120,000

Figure 1: Example milestone recommendation for a new campaign.

3.2 Backtesting

The proposed recommendation method was evaluated by comparing the generated milestone ladders with the historical milestone configurations. The evaluation measured differences in milestone thresholds and payout amounts, while also checking whether the recommended payouts remained within the

allocated campaign budgets.

Backtesting Summary:

	Metric	Value
0	Average Threshold Difference	96940.88
1	Max Threshold Difference	1066988
2	Average Payout Difference	0.0
3	Max Payout Difference	0
4	Campaigns Within Budget	50 / 50
5	Avg Milestones Reached per Post	2.16 / 5

Figure 2: Summary of backtesting results.

The evaluation demonstrates strong performance across evaluated campaigns, with **100%** of recommendations remaining strictly within the allocated campaign budget. These results confirm that the proposed methodology produces recommendations that are consistent with historical campaign performance while guaranteeing budget compliance.